INTRODUCTION

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I conducted a data analysis project focused on predicting customer churn for SyriaTel.

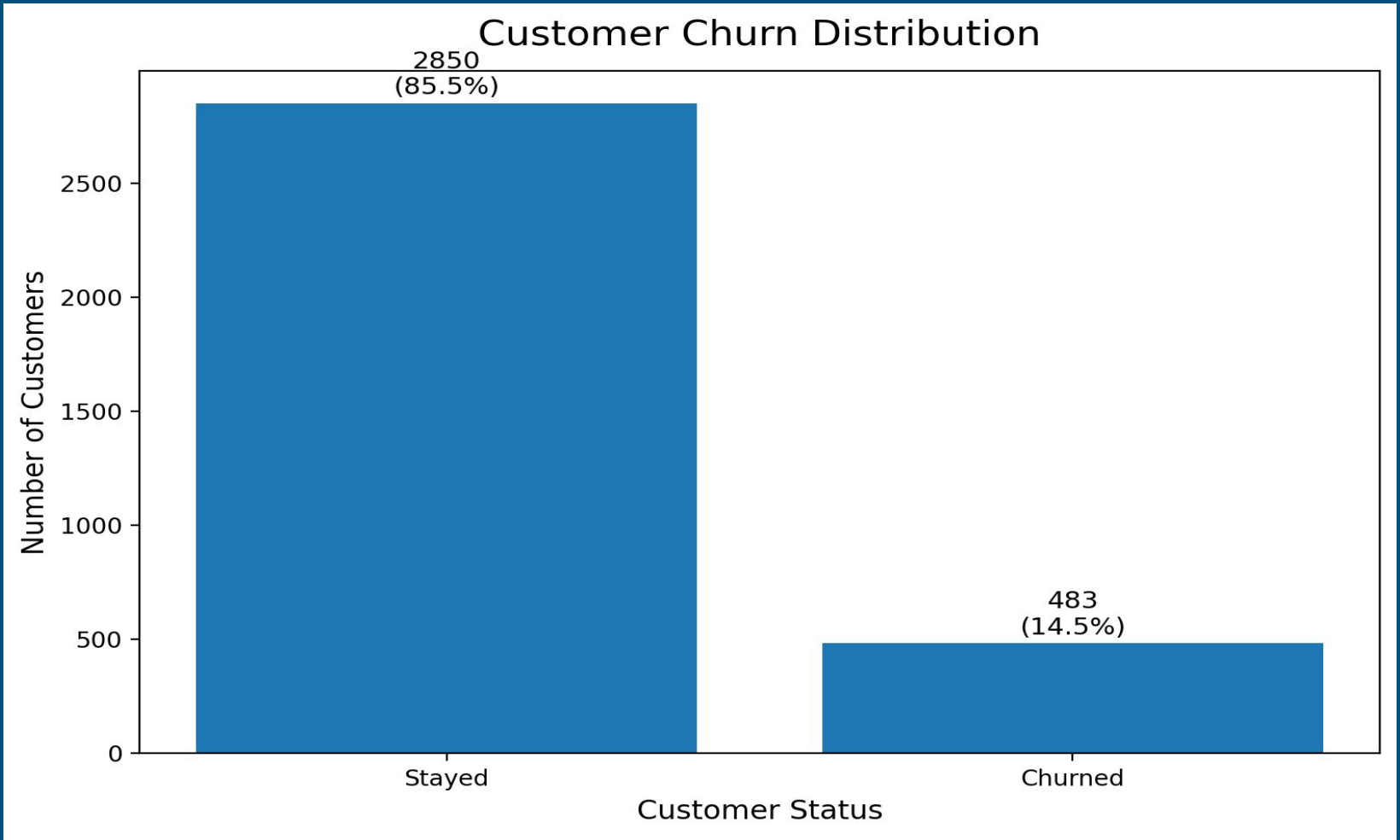
This work involved applying machine learning techniques to generate insights and actionable recommendations for customer retention.

SYRIATEL CUSTOMER CHURN ANALYSIS

- Reducing Customer Loss Through Data-Driven Insights.
- Prepared for: Customer Retention Team.

BUSINESS PROBLEM

- Customer churn leads to revenue loss.
- Need to identify customers likely to leave early.
- Enable proactive retention strategies.



This chart shows that most customers stayed but a significant portion still left the company. It confirms that churn is a real business problem worth addressing early.

WHY THIS MATTERS

- Acquiring new customers is more expensive than retaining existing ones.
- Early identification allows targeted intervention.
- Improves customer satisfaction and loyalty.

DATA OVERVIEW

- 3,333 customer records analyzed.
- Includes usage patterns, service plans, and customer interactions.
- Goal: Find patterns linked to churn.

APPROACH

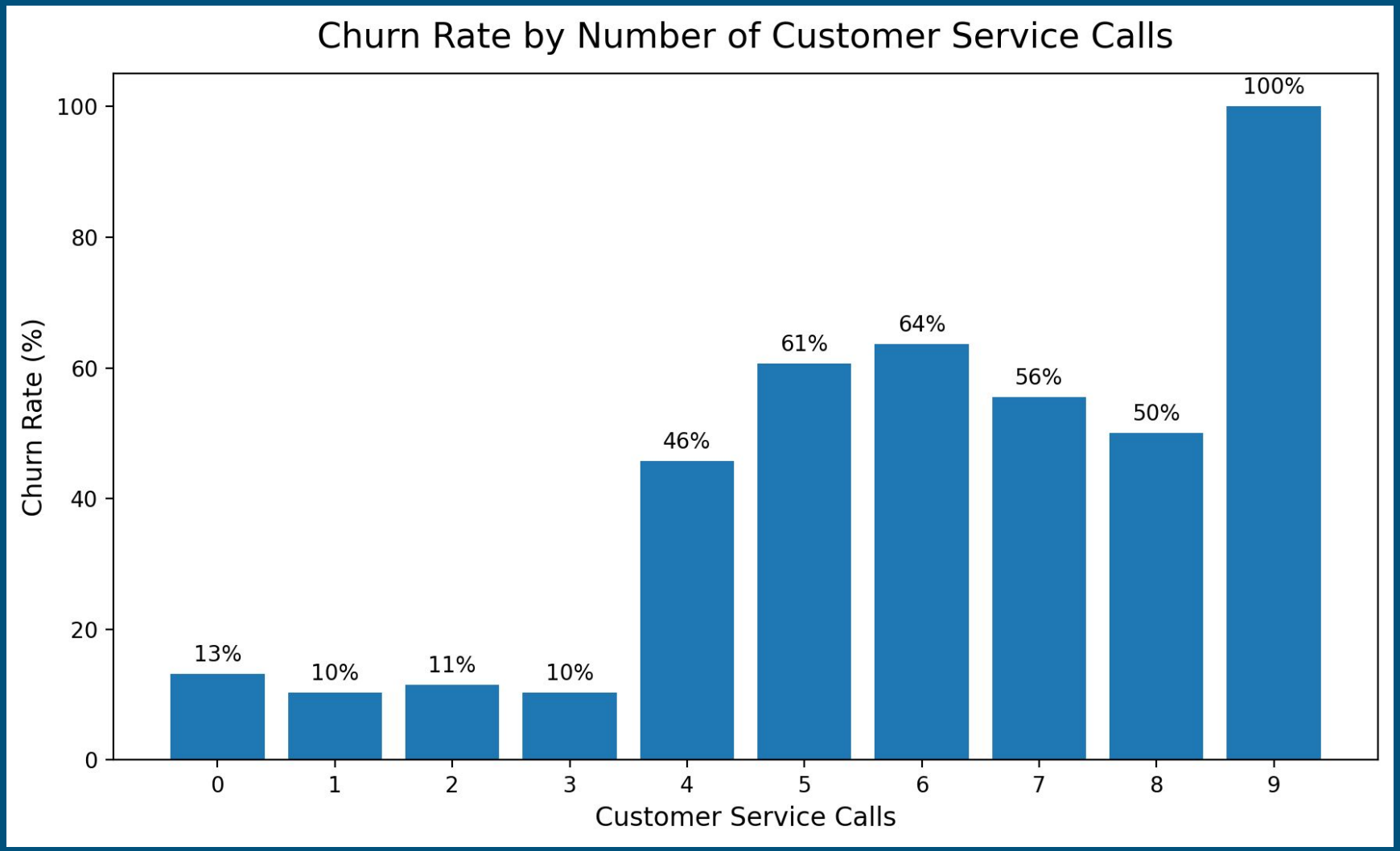
- Used classification models to predict churn.
- Started simple, then improved models iteratively.
- Compared multiple models to find best performance.

KEY METRIC(RECALL)

- Focus: Catch as many potential churners as possible.
- Higher recall = fewer missed at-risk customers.
- Some false alarms are acceptable.

KEY FINDINGS

- Frequent customer service calls increase churn risk.
- Heavy users are more sensitive to service quality.
- Plan types influence customer behavior.



Customers who contact customer service more times are much more likely to leave. This suggests that service experience is one of the strongest warning signs of churn.

BUSINESS RECOMMENDATIONS

- Prioritize customers with many service calls.
- Improve customer support experience.
- Offer targeted retention incentives.
- Monitor high-usage customers.

NEXT STEPS

- Deploy model into operations.
- Continuously update with new data.
- Test and refine retention strategies.

THANK YOU

- Questions and discussion?